

The Laplace transform Solutions to Problem Sheet 2

1. Using the table of Laplace transforms, we obtain the following.

(a) $\mathcal{L}\{2e^{4t}\} = \frac{2}{s-4}$ for $s > 4$.

(b) $\mathcal{L}\{3e^{-2t}\} = \frac{3}{s+2}$ for $s > -2$.

(c) $\mathcal{L}\{5t - 3\} = \frac{5}{s^2} - \frac{3}{s}$ for $s > 0$.

(d) $\mathcal{L}\{2t^2 - 3\} = 2\frac{2!}{s^3} - \frac{3}{s} = \frac{4}{s^3} - \frac{3}{s}$ for $s > 0$.

(e) $\mathcal{L}\{3\cos(5t)\} = \frac{3s}{s^2 + 25}$ for $s > 0$.

(f) $\mathcal{L}\{10\sin(6t)\} = 10\left(\frac{6}{s^2 + 36}\right) = \frac{60}{s^2 + 36}$ for $s > 0$.

2. $F(t) = 4H(t-2)$. Hence, $\bar{F}(s) = \frac{4e^{-2s}}{s}$.

3. Using the table of Laplace transforms, we have

$$\mathcal{L}\{\sinh(kt)\} = \frac{k}{s^2 - k^2}.$$

Combining this with the First Shifting Theorem,

$$\mathcal{L}\{e^{\alpha t}f(t)\} = \bar{f}(s - \alpha),$$

we obtain

$$\mathcal{L}\{e^{\alpha s} \sinh(kt)\} = \frac{k}{(s - \alpha)^2 - k^2}.$$

4. Using the table of Laplace transforms, we obtain the following.

(a) $\bar{f}(s) = \frac{1}{s+1} = \frac{1}{s - (-1)}$, so $f(t) = e^{-t}$.

(b) $\bar{f}(s) = \frac{1}{(s-3)^2} = \frac{1}{p^2}$, where $p = s - 3$. Hence, using $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$ and the First Shifting Theorem, $f(t) = te^{3t}$.

(c) $\bar{f}(s) = \frac{a}{s(s+a)}$. Using partial fractions,

$$\frac{a}{s(s+a)} = \frac{1}{s} - \frac{1}{s+a}.$$

Hence, $f(t) = 1 - e^{-at}$.

(d) $\bar{f}(s) = \frac{k^2}{s(s^2 + k^2)}$. Using partial fractions,

$$\begin{aligned}\frac{k^2}{s(s^2 + k^2)} &= \frac{A}{s} + \frac{Bs + C}{s^2 + k^2}, \\ &= \frac{As^2 + Ak^2 + Bs^2 + Cs}{s(s^2 + k^2)}, \\ &= \frac{(A + B)s^2 + Cs + Ak^2}{s(s^2 + k^2)}, \quad \text{i.e. } A = 1, B = -1, C = 0, \\ &= \frac{1}{s} - \frac{s}{s^2 + k^2}.\end{aligned}$$

Hence,

$$f(t) = 1 - \cos(kt).$$

(e) $\bar{f}(s) = \frac{6s - 4}{s^2 - 4s + 20}$. Completing the square of the denominator and forcing the term $(s - 2)$ to appear in the numerator,

$$\begin{aligned}\frac{6s - 4}{s^2 - 4s + 20} &= \frac{6s - 4}{(s - 2)^2 + 16}, \\ &= \frac{6(s - 2) + 8}{(s - 2)^2 + 16}, \\ &= \frac{6(s - 2)}{(s - 2)^2 + 16} + \frac{8}{(s - 2)^2 + 16}.\end{aligned}$$

Using the First Shifting Theorem,

$$f(t) = 6e^{2t} \cos(4t) + 2e^{2t} \sin(4t).$$

(f) $\bar{f}(s) = \frac{e^{-2s}}{s^4}$. Using $\mathcal{L}^{-1}\left\{\frac{1}{s^4}\right\} = \frac{t^3}{3!} = \frac{t^3}{6}$ and the Second Shifting Theorem,

$$f(t) = \frac{1}{6}(t - 2)^3 H(t - 2),$$

i.e.

$$f(t) = \begin{cases} 0 & \text{for } 0 \leq t \leq 2, \\ \frac{1}{6}(t - 2)^3 & \text{for } t > 2. \end{cases}$$

5. (a)

$$\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + 5y = 0, \quad \text{with } y(0) = 1, y'(0) = -3.$$

Take the Laplace transform of both sides of the equation and solving for $\bar{y}$,

$$\mathcal{L}\{y''\} - 6\mathcal{L}\{y'\} + 5\mathcal{L}\{y\} = 0.$$

$$[s^2\bar{y} - sy(0) - y'(0)] - 6[s\bar{y} - y(0)] + 5\bar{y} = 0.$$

$$s^2\bar{y} - s + 3 - 6s\bar{y} + 6 + 5\bar{y} = 0.$$

$$(s^2 - 6s + 5)\bar{y} = s - 9.$$

$$\bar{y} = \frac{s - 9}{s^2 - 6s + 5}.$$

To invert the Laplace transform, we use partial fractions:

$$\begin{aligned}\bar{y} &= \frac{s - 9}{s^2 - 6s + 5} = \frac{s - 9}{(s - 1)(s - 5)}, \\ &= \frac{A}{s - 1} + \frac{B}{s - 5}, \\ &= \frac{A(s - 5) + B(s - 1)}{(s - 1)(s - 5)}, \\ &= \frac{(A + B)s - (5A + B)}{(s - 1)(s - 5)}, \quad \text{i.e. } A = 2, B = -1,\end{aligned}$$

Hence,

$$\bar{y} = \frac{2}{s - 1} - \frac{1}{s - 5}.$$

Using the table of Laplace transforms,

$$y = 2e^t - e^{5t}.$$

(b)

$$\frac{d^2y}{dt^2} - 3\frac{dy}{dt} + 2y = e^{3t}, \quad \text{with } y(0) = 1, y'(0) = 0.$$

Take the Laplace transform of both sides and solve for $\bar{y}$,

$$\mathcal{L}\{y''\} - 3\mathcal{L}\{y'\} + 2\mathcal{L}\{y\} = \mathcal{L}\{e^{3t}\}.$$

$$[s^2\bar{y} - sy(0) - y'(0)] - 3[s\bar{y} - y(0)] + 2\bar{y} = \frac{1}{s - 3}.$$

$$s^2\bar{y} - s - 3s\bar{y} + 3 + 2\bar{y} = \frac{1}{s - 3}.$$

$$(s^2 - 3s + 2)\bar{y} = s - 3 + \frac{1}{s - 3} = \frac{(s - 3)^2 + 1}{s - 3}.$$

$$\bar{y} = \frac{s^2 - 6s + 10}{(s^2 - 3s + 2)(s - 3)} = \frac{s^2 - 6s + 10}{(s - 1)(s - 2)(s - 3)}.$$

Using partial fractions,

$$\begin{aligned}\frac{s^2 - 6s + 10}{(s - 1)(s - 2)(s - 3)} &= \frac{A}{s - 1} + \frac{B}{s - 2} + \frac{C}{s - 3}, \\ &= \frac{A(s - 2)(s - 3) + B(s - 1)(s - 3) + C(s - 1)(s - 2)}{(s - 1)(s - 2)(s - 3)}, \\ &= \frac{5}{2(s - 1)} - \frac{2}{s - 2} + \frac{1}{2(s - 3)}.\end{aligned}$$

Hence, using the table of Laplace transforms,

$$f(t) = \frac{5}{2}e^t - 2e^{2t} + \frac{1}{2}e^{3t}.$$

(c)

$$\frac{dy}{dt} - 6y = 2\sin(3t), \quad \text{with } y(0) = -1.$$

Taking the Laplace transform of both sides and solving for $\bar{y}$,

$$\mathcal{L}\{y'\} - 6\mathcal{L}\{y\} = 2\mathcal{L}\{\sin(3t)\}.$$

$$s\bar{y} - y(0) - 6\bar{y} = \frac{6}{s^2 + 9}.$$

$$s\bar{y} + 1 - 6\bar{y} = \frac{6}{s^2 + 9}.$$

$$(s - 6)\bar{y} = \frac{6}{s^2 + 9} - 1.$$

$$\bar{y} = \frac{6}{(s^2 + 9)(s - 6)} - \frac{1}{s - 6} = \frac{-s^2 - 3}{(s^2 + 9)(s - 6)}.$$

Using partial fractions,

$$\begin{aligned} \frac{-s^2 - 3}{(s^2 + 9)(s - 6)} &= \frac{As + B}{s^2 + 9} + \frac{C}{s - 6}, \\ &= \frac{(A + C)s^2 + (B - 6A)s + 9C - 6B}{(s^2 + 9)(s - 6)}, \end{aligned}$$

Hence,

$$A + C = -1, \quad B - 6A = 0, \quad 9C - 6B = -3.$$

Solving simultaneously (e.g. using Gaussian elimination)

$$A = -\frac{2}{15}, \quad B = -\frac{4}{5}, \quad C = -\frac{13}{15}.$$

Hence,

$$\bar{f}(s) = -\frac{2s}{15(s^2 + 9)} - \frac{4}{5(s^2 + 9)} - \frac{13}{15(s - 6)}.$$

Using the table of Laplace transforms, and observing that

$$\frac{4}{5(s^2 + 9)} = \frac{4(3)}{15(s^2 + 9)},$$

we obtain

$$f(t) = -\frac{2}{15} \cos(3t) - \frac{4}{15} \sin(3t) - \frac{13}{15} e^{6t}.$$

6.

$$\frac{d^2y}{dt^2} - 2a\frac{dy}{dt} + (a^2 + b^2)y = 0, \quad \text{with } y(0) = 0, \quad y'(0) = 1.$$

Taking the Laplace transform of both sides and solving for $\bar{y}$,

$$\mathcal{L}\{y''\} - 2a\mathcal{L}\{y'\} + (a^2 + b^2)\mathcal{L}\{y\} = 0.$$

$$[s^2\bar{y} - sy(0) - y'(0)] - 2a[s\bar{y} - y(0)] + (a^2 + b^2)\bar{y} = 0.$$

$$s^2\bar{y} - 1 - 2as\bar{y} + (a^2 + b^2)\bar{y} = 0.$$

$$(s^2 - 2as + a^2 + b^2)\bar{y} = 1.$$

$$\bar{y} = \frac{1}{s^2 - 2as + a^2 + b^2} = \frac{1}{(s - a)^2 + b^2}.$$

Using $\mathcal{L}^{-1}\left\{\frac{1}{s^2 + b^2}\right\} = \frac{1}{b}\sin(bt)$ and the First Shifting Theorem,

$$f(t) = \frac{1}{b}e^{at}\sin(bt).$$

7.

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = H(t) - H(t - 4), \quad \text{with } y(0) = 0, \quad y'(0) = 0.$$

Taking the Laplace transform of both sides and solving for $\bar{y}$,

$$s^2\bar{y} + 4s\bar{y} + 3\bar{y} = \frac{1}{s} - \frac{e^{-4s}}{s}.$$

$$(s^2 + 4s + 3)\bar{y} = \frac{1}{s} - \frac{e^{-4s}}{s}.$$

$$\bar{y} = \frac{1 - e^{-4s}}{s(s^2 + 4s + 3)} = \frac{1 - e^{-4s}}{s(s + 3)(s + 1)}.$$

Using partial fractions,

$$\begin{aligned} \frac{1}{s(s + 3)(s + 1)} &= \frac{A}{s} + \frac{B}{s + 3} + \frac{C}{s + 1}, \\ &= \frac{A(s + 3)(s + 1) + Bs(s + 1) + Cs(s + 3)}{s(s + 3)(s + 1)}, \\ &= \frac{1}{3s} + \frac{1}{6(s + 3)} - \frac{1}{2(s + 1)}. \end{aligned}$$

Hence,

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2 + 4s + 3)}\right\} = \frac{1}{3} + \frac{1}{6}e^{-3t} - \frac{1}{2}e^{-t}.$$

Using the Second Shifting Theorem,

$$\mathcal{L}^{-1}\left\{\frac{e^{-4s}}{s(s^2 + 4s + 3)}\right\} = \left(\frac{1}{3} + \frac{1}{6}e^{-3(t-4)} - \frac{1}{2}e^{-(t-4)}\right)H(t - 4).$$

Finally,

$$f(t) = \frac{1}{3} + \frac{1}{6}e^{-3t} - \frac{1}{2}e^{-t} - \left(\frac{1}{3} + \frac{1}{6}e^{-3(t-4)} - \frac{1}{2}e^{-(t-4)}\right)H(t - 4).$$

8. (a) According to the table, we have

$$\mathcal{L}\{\sin t\} = \frac{1}{s^2 + 1} \text{ and } \mathcal{L}\{e^{-2t}f(t)\} = \bar{f}(s+2) \text{ (First Shifting Theorem).}$$

Hence,

$$\mathcal{L}\{e^{-2t} \sin t\} = \frac{1}{(s+2)^2 + 1} = \frac{1}{s^2 + 4s + 5}.$$

(b) According to the table, we have

$$\mathcal{L}\{1\} = \frac{1}{s}, \quad \mathcal{L}\{\cos t\} = \frac{s}{s^2 + 1}.$$

The First Shifting Theorem gives

$$\mathcal{L}\{e^{-2t} \cos t\} = \frac{s+2}{(s+2)^2 + 1} = \frac{s+2}{s^2 + 4s + 5}.$$

Combining with the result from part (a), we have

$$\begin{aligned} \mathcal{L}\left\{\frac{1}{5}[1 - e^{-2t} \cos t - 2e^{-2t} \sin t]\right\} &= \frac{1}{5} [\mathcal{L}\{1\} - \mathcal{L}\{e^{-2t} \cos t\} - 2\mathcal{L}\{e^{-2t} \sin t\}], \\ &= \frac{1}{5} \left[\frac{1}{s} - \frac{s+2}{s^2 + 4s + 5} - \frac{2}{s^2 + 4s + 5} \right], \\ &= \frac{1}{5} \left[\frac{s^2 + 4s + 5 - s^2 - 2s - 2s}{s(s^2 + 4s + 5)} \right], \\ &= \frac{1}{s(s^2 + 4s + 5)}. \end{aligned}$$

(c) Let $v = t - 3$. Using the result from (b) and the Second Shifting Theorem,

$$\mathcal{L}\left\{\frac{1}{5}[1 - e^{-2v} \cos v - 2e^{-2v} \sin v]H(v)\right\} = \frac{e^{-3s}}{s(s^2 + 4s + 5)}.$$

We have the ODE

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 5y = H(t-3), \quad \text{with } y(0) = 0, \quad y'(0) = 1.$$

Taking the Laplace transform of both sides,

$$\begin{aligned} [s^2\bar{y} - sy(0) - y'(0)] + 4[s\bar{y} - y(0)] + 5\bar{y} &= \frac{e^{-3s}}{s}, \\ \Rightarrow (s^2 + 4s + 5)\bar{y} - 1 &= \frac{e^{-3s}}{s}, \\ \Rightarrow \bar{y} &= \frac{e^{-3s}}{s(s^2 + 4s + 5)} + \frac{1}{s^2 + 4s + 5}. \end{aligned}$$

Inverting $\bar{y}$ using the results in (a) and (c),

$$y(t) = \frac{1}{5} [1 - e^{-2(t-3)} \cos(t-3) - 2e^{-2(t-3)} \sin(t-3)] H(t-3) + e^{-2t} \sin t.$$

9. The Final Value Theorem states that

$$\lim_{t \rightarrow \infty} f(t) = \lim_{s \rightarrow 0} [s\bar{f}(s)].$$

Let $f(t) = 1 - e^{-t}$. Firstly, since $\lim_{t \rightarrow \infty} e^{-t} = 0$, we have

$$\lim_{t \rightarrow \infty} f(t) = 1 - 0 = 1.$$

Taking the Laplace transform,

$$\mathcal{L}\{f(t)\} = \bar{f}(s) = \frac{1}{s} - \frac{1}{s+1} = \frac{1}{s(s+1)}.$$

Hence,

$$\lim_{s \rightarrow 0} [s\bar{f}(s)] = \lim_{s \rightarrow 0} \left[\frac{1}{s+1} \right] = 1.$$

Hence, the Final Value Theorem applies here.

10. (a) Let $f(t) = \cos(at)$. Using the definition of the Laplace transform,

$$\bar{f}(s) = \int_0^{\infty} e^{-st} \cos(at) dt.$$

Integrating by parts twice,

$$\begin{aligned} \bar{f}(s) &= \underbrace{\left[\frac{1}{a} e^{-st} \sin(at) \right]_0^{\infty}}_{=0} + \frac{s}{a} \int_0^{\infty} e^{-st} \sin(at) dt, \\ &= \frac{s}{a} \underbrace{\left[-\frac{1}{a} e^{-st} \cos(at) \right]_0^{\infty}}_{=1/a} - \frac{s^2}{a^2} \underbrace{\int_0^{\infty} e^{-st} \cos(at) dt}_{=\bar{f}(s)}, \\ &= \frac{s}{a^2} - \frac{s^2}{a^2} \bar{f}(s). \end{aligned}$$

Rearranging,

$$\begin{aligned} \bar{f}(s) \left(1 + \frac{s^2}{a^2} \right) &= \frac{s}{a^2}, \\ \Rightarrow \bar{f}(s) \left(\frac{s^2 + a^2}{a^2} \right) &= \frac{s}{a^2}, \\ \Rightarrow \bar{f}(s) &= \frac{s}{s^2 + a^2}. \end{aligned}$$

(b) Let $f(t) = \sinh(at)$. Then

$$\bar{f}(s) = \int_0^{\infty} e^{-st} \sinh(at) dt.$$

Integrating by parts twice (using the hint to differentiate/integrate $\sinh(at)$ and $\cosh(at)$),

$$\begin{aligned}
 \bar{f}(s) &= \underbrace{\left[-\frac{1}{s} e^{-st} \sinh(at) \right]_0^\infty}_{=0} + \frac{a}{s} \int_0^\infty e^{-st} \cosh(at) dt, \\
 &= \frac{a}{s} \underbrace{\left[-\frac{1}{s} e^{-st} \cosh(at) \right]_0^\infty}_{=1/s} + \frac{a^2}{s^2} \underbrace{\int_0^\infty e^{-st} \sinh(at) dt}_{=\bar{f}(s)}, \\
 &= \frac{a}{s^2} + \frac{a^2}{s^2} \bar{f}(s).
 \end{aligned}$$

Rearranging,

$$\begin{aligned}
 \bar{f}(s) \left(1 - \frac{a^2}{s^2} \right) &= \frac{a}{s^2}, \\
 \Rightarrow \bar{f}(s) \left(\frac{s^2 - a^2}{s^2} \right) &= \frac{a}{s^2}, \\
 \Rightarrow \bar{f}(s) &= \frac{a}{s^2 - a^2}.
 \end{aligned}$$

(c) Let $f(t) = \cosh(at)$. Then

$$\bar{f}(s) = \int_0^\infty e^{-st} \cosh(at) dt.$$

Integrating by parts twice,

$$\begin{aligned}
 \bar{f}(s) &= \underbrace{\left[-\frac{1}{s} e^{-st} \cosh(at) \right]_0^\infty}_{=1/s} + \frac{a}{s} \int_0^\infty e^{-st} \sinh(at) dt, \\
 &= \frac{1}{s} + \frac{a}{s} \underbrace{\left[-\frac{1}{s} e^{-st} \sinh(at) \right]_0^\infty}_{=0} + \frac{a^2}{s^2} \underbrace{\int_0^\infty e^{-st} \cosh(at) dt}_{=\bar{f}(s)}, \\
 &= \frac{1}{s} + \frac{a^2}{s^2} \bar{f}(s).
 \end{aligned}$$

Rearranging,

$$\begin{aligned}
 \bar{f}(s) \left(1 - \frac{a^2}{s^2} \right) &= \frac{1}{s}, \\
 \Rightarrow \bar{f}(s) \left(\frac{s^2 - a^2}{s^2} \right) &= \frac{1}{s}, \\
 \Rightarrow \bar{f}(s) &= \frac{s}{s^2 - a^2}.
 \end{aligned}$$

(d) Let $f(t) = t \sin(kt)$. Then

$$\bar{f}(s) = \int_0^\infty t e^{-st} \sin(kt) dt.$$

Integrate by parts, setting

$$\begin{aligned} u &= te^{-st}, & v' &= \sin(kt), \\ u' &= (1-st)e^{-st} \text{ (product rule)}, & v &= -\frac{1}{k} \cos(kt). \end{aligned}$$

Then

$$\bar{f}(s) = \underbrace{\left[-\frac{t}{k} e^{-st} \cos(kt) \right]_0^\infty}_{=0} + \frac{1}{k} \int_0^\infty (1-st)e^{-st} \cos(kt) dt.$$

Integrating by parts again, setting

$$\begin{aligned} u &= (1-st)e^{-st}, & v' &= \cos(kt), \\ u' &= (s^2t-2s)e^{-st}, & v &= \frac{1}{k} \sin(kt). \end{aligned}$$

$$\begin{aligned} \bar{f}(s) &= \frac{1}{k^2} \underbrace{\left[(1-st)e^{st} \sin(kt) \right]_0^\infty}_{=0} - \frac{s}{k^2} \int_0^\infty (st-2)e^{-st} \sin(kt) dt, \\ &= -\frac{s^2}{k^2} \underbrace{\int_0^\infty te^{-st} \sin(kt) dt}_{= \bar{f}(s)} + \frac{2s}{k^2} \underbrace{\int_0^\infty e^{-st} \sin(kt) dt}_{= \mathcal{L}\{\sin(kt)\}}. \\ &= -\frac{s^2}{k^2} \bar{f}(s) + \frac{2s}{k^2} \left(\frac{k}{s^2 + k^2} \right), \end{aligned}$$

Where we use the table to take the Laplace transform of $\sin(kt)$. Rearranging,

$$\begin{aligned} \bar{f}(s) \left(1 + \frac{s^2}{k^2} \right) &= \frac{2ks}{k^2(s^2 + k^2)}, \\ \Rightarrow \bar{f}(s) \left(\frac{s^2 + k^2}{k^2} \right) &= \frac{2ks}{k^2(s^2 + k^2)}, \\ \Rightarrow \bar{f}(s) &= \frac{2ks}{(s^2 + k^2)^2}. \end{aligned}$$

(e) Let $f(t) = t \cos(kt)$. Then

$$\bar{f}(s) = \int_0^\infty te^{-st} \cos(kt) dt.$$

Integrate by parts, setting

$$\begin{aligned} u &= te^{-st}, & v' &= \cos(kt), \\ u' &= (1-st)e^{-st}, & v &= \frac{1}{k} \sin(kt). \end{aligned}$$

Then

$$\bar{f}(s) = \underbrace{\left[\frac{t}{k} e^{-st} \sin(kt) \right]_0^\infty}_{=0} - \frac{1}{k} \int_0^\infty (1-st)e^{-st} \sin(kt) dt.$$

Integrating by parts again, setting

$$\begin{aligned} u &= (1 - st)e^{-st}, & v' &= \sin(kt), \\ u' &= (s^2t - 2s)e^{-st}, & v &= -\frac{1}{k} \cos(kt). \end{aligned}$$

$$\begin{aligned} \bar{f}(s) &= \frac{1}{k^2} \underbrace{\left[(1 - st)e^{-st} \cos(kt) \right]_0^\infty}_{=-1} - \frac{s}{k^2} \int_0^\infty (st - 2)e^{-st} \cos(kt) \, dt, \\ &= -\frac{1}{k^2} - \frac{s^2}{k^2} \underbrace{\int_0^\infty te^{-st} \cos(kt) \, dt}_{=\bar{f}(s)} + \frac{2s}{k^2} \underbrace{\int_0^\infty e^{-st} \cos(kt) \, dt}_{=\mathcal{L}\{\cos(kt)\}}, \\ &= -\frac{1}{k^2} - \frac{s^2}{k^2} \bar{f}(s) + \frac{2s}{k^2} \left(\frac{s}{s^2 + k^2} \right), \end{aligned}$$

where we use the table to take the Laplace transform of $\cos(kt)$. Rearranging,

$$\begin{aligned} \bar{f}(s) \left(1 + \frac{s^2}{k^2} \right) &= \frac{1}{k^2} \left(-1 + \frac{2s^2}{s^2 + k^2} \right), \\ \Rightarrow \bar{f}(s) \left(\frac{s^2 + k^2}{k^2} \right) &= \frac{1}{k^2} \left(\frac{s^2 - k^2}{s^2 + k^2} \right), \\ \Rightarrow \bar{f}(s) &= \frac{s^2 - k^2}{(s^2 + k^2)^2}. \end{aligned}$$